

Question 1:

$$f_n(x) = \begin{cases} \sin(\pi x), & \text{if } 2n \leq x < (2n+1), \\ 0, & \text{otherwise.} \end{cases}$$

a)**b)**

To test of a pointwise convergence of $f_n(x)$ to $f(x)$, we need to consider different cases for x .

Case 1: $x \in [0, 1)$

$$f_n(x) = \sin(\pi x) \text{ for all } n \geq 0$$

$$\lim_{n \rightarrow \infty} f_n(x) = \sin(\pi x)$$

$$\therefore f(x) = \sin(\pi x)$$

Case 2: $x \in [1, 2)$

$$f_n(x) = 0 \text{ for all } n \geq 0$$

$$\lim_{n \rightarrow \infty} f_n(x) = 0$$

$$\therefore f(x) = 0$$

Case 3: $x \in [2, 3)$

$$f_n(x) = \sin(\pi x) \text{ for all } n \geq 1$$

$$\lim_{n \rightarrow \infty} f_n(x) = \sin(\pi x)$$

$$\therefore f(x) = \sin(\pi x)$$

Case 4: $x \in [3, 4)$

$$f_n(x) = 0 \text{ for all } n \geq 1$$

$$\lim_{n \rightarrow \infty} f_n(x) = 0$$

$$\therefore f(x) = 0$$

$\vdots$

Case k: $x \in [k, k+1)$

$$f_n(x) = \begin{cases} \sin(\pi x), & \text{if } k \text{ is even and } n \geq k/2, \\ 0, & \text{if } k \text{ is odd and } n \geq (k-1)/2. \end{cases}$$

$$\lim_{n \rightarrow \infty} f_n(x) = \begin{cases} \sin(\pi x), & \text{if } k \text{ is even,} \\ 0, & \text{if } k \text{ is odd.} \end{cases}$$

$$\therefore f(x) = \begin{cases} \sin(\pi x), & \text{if } k \text{ is even,} \\ 0, & \text{if } k \text{ is odd.} \end{cases}$$

Thus we have that the pointwise limit function $f(x)$ is given by;

$$f(x) = \begin{cases} \sin(\pi x), & \text{if } \lfloor x \rfloor \text{ is even,} \\ 0, & \text{if } \lfloor x \rfloor \text{ is odd.} \end{cases}$$

c)